

Maanik Gogna

4th Year Computer Engineering | University of Toronto

Education

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University of Toronto

Sept 2021 - Apr 2026

BASc Computer Engineering: Minor in Artificial Intelligence + PEY Co-op

Relevant Courses: Computer Architecture, Operating Systems, Data Structures and Algorithms, Digital Systems

Work Experiences

Silicon Validation Intern

May 2024 - Aug 2025

Alphawave SEMI (acquired by Qualcomm)

- Managed multi-project regression flows and improved bit error rate (BER) by 2-3 orders of magnitude on critically affected projects through iterative research and analysis, ensuring customer and industry compliance
- Collaborated with a team of senior engineers to validate 5 silicon projects, improving transmitter eye jitter and BER performance across multiple data rates and test conditions for PCIe and Ethernet standards
- Redesigned the PCIe Interoperability testing infrastructure, enabling integration into the main software environment for automated, scalable testing
- Increased test coverage by 20% by developing new test cases targeting silicon weak points and edge cases
- Improved manual validation processing time by 25% through the development of automated data processing tools, featuring an integrated back-end query service and front-end user interface

Technical Design Experiences

Capstone Team Member

May 2025 - Present

MannLab

- Leading team of 4 under Professor Steve Mann to implement augmented reality glasses for visually impaired users, improving navigation through obstacle detection and multimodal feedback (haptic, audio, and visual)
- Developing machine learning model to predict collisions and suggest avoidance actions in less than 200 ms
- Designing facial recognition and

Electronics and Electrical Systems Lead

May 2023 - Present

University of Toronto Supermileage Team - Prototype, Urban Concept Cars

- Design printed circuit board (PCB) for successive designs on Prototype and Urban Concept cars to implement 48V battery management system and self-designed telemetry system for hydrogen car
- Drove electrical systems development for team's electric and hydrogen cars participating in annual Shell Eco-Marathon Americas Competition

Personal Projects

SafeStride - Full Stack Web App | [GitHub](#)

May 2023 - Present

- Developed wearable device to improve navigational safety of visually impaired users by integrating obstacle detection and directional haptic and audio feedback alerts
- Implemented real-time obstacle detection alerts with OpenCV and YOLOv8 model within 100 ms constraint by implementing Flask communication link between back-end machine learning server and two mobile devices

VacayAway - Full Stack Web App | [GitHub](#)

ReactJS, Firebase, Google Cloud, Flask, Scikit-learn

- Designed full stack ReactJS trip planner and scheduler that supports 50+ international cities
- Leveraged Scikit-learn machine learning on Flask server to efficiently cluster together attractions by location
- Complete start-to-finish trip generation and scheduling (flight, hotel, restaurants, attractions) in <1 minute

Technical Skills & Interests

Languages: Python | C++ | C | Swift | Java | Javascript | Verilog | ARM Assembly

Hardware: Intel Quartus Prime | ModelSim | Altium Designer | DE1-SoC Board | Arduino

Other: Git | Linux | Unix | React | NumPy | pandas | scikit-learn | CoreML | Flask | MongoDB | MATLAB

Interests: Chess | Reading | Pop Music | Hiking | Outdoor Exploration | Board Games